

Route decision: Port of Shanghai → Port of Rotterdam

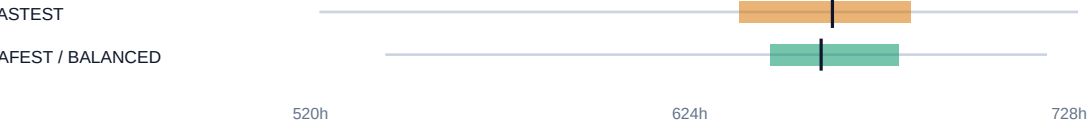
PORT-SHANGHAI → PORT-ROTTERDAM · mode sea (STRICT) · cargo general · priority normal · scenarios: none · planned 2026-09-29 13:47 UTC

Recommendation: SAFEST / BALANCED, SEA via Malacca, Babel, Suez, Gibral. Expected arrival in **643 h** (26.8 days); plan for 657 h (p85), worst case 679 h (p95). Cost: freight S\$3,979. Peak leg risk 8%.

Options

Option	Mode	Via	Plan h	p50 h	p85 h	p95 h	Freight	Risk
FASTEST	SEA	Malacca, Babel, Suez, Gibral	520	635	660	682	S\$3,486	18%
SAFEST / BALANCED	SEA	Malacca, Babel, Suez, Gibral	538	643	657	679	S\$3,979	8%

ETA bands (bar p50–p95, tick p85, line plan → fully-correlated worst case)



Routes



Recommended route solid, alternatives dashed. Blue sea, amber air, green rail, orange road.

FASTEST: SEA via Malacca, Babel, Suez, Gibral

SEA via Malacca, Babel, Suez, Gibral; expected 635.0 h (p85 660.5 h, p95 682.1 h), S\$3,486 freight, 0 handoffs. 8.4 h faster than the next option. S\$491.90 cheaper (freight) than the lowest-cost alternative.

#	Mode	To	km	Hours	Delay p85	Cost	Risk	Signal
1	SEA	Strait of Malacca	3,810	108.9	+26.6	S\$730.22	0%	
2	SEA	Bab el-Mandeb Strait	6,585	188.1	+27.5	S\$1,262	0%	
3	SEA	Suez Canal	2,302	65.8	+28.5	S\$441.22	0%	
4	SEA	Strait of Gibraltar	3,548	101.4	+20.6	S\$679.99	0%	
5	SEA	Port of Rotterdam	1,945	55.6	+55.3	S\$372.72	18%	LIVE_NEWS: Temporary transformer in Europort relieves grid congestion - Port of Rotterdam

Transit 519.7 h + transfers 0.0 h + announced scenario delay 0.0 h = plan 519.7 h. Model delay buffer p50/p85/p95: +115.3 / +158.6 / +208.2 h. Route band p50 635.0 h, p85 660.5 h, p95 682.1 h (independent legs); 727.9 h if all legs are late together. Costs: linehaul S\$3,486, transfers S\$0.00, disruption surcharge S\$0.00.

Why the slowest leg (CHOKE-GIBRAL → PORT-ROTTERDAM, p85 55.3 h) is slow: exact Shapley attribution of the p85 delay model, hours relative to a 28.9 h baseline



SAFEST / BALANCED: SEA via Malacca, Babel, Suez, Gibral

SEA via Malacca, Babel, Suez, Gibral; expected 643.4 h (p85 657.4 h, p95 678.6 h), S\$3,979 freight, 2 handoffs. 8.4 h slower than FASTEST. S\$491.90 more (freight) than FASTEST.

#	Mode	To	km	Hours	Delay p85	Cost	Risk	Signal
1	SEA	Strait of Malacca	3,810	108.9	+26.6	S\$730.22	0%	
2	SEA	Bab el-Mandeb Strait	6,585	188.1	+27.5	S\$1,262	0%	
3	SEA	Suez Canal	2,302	65.8	+28.5	S\$441.22	0%	
4	SEA	Strait of Gibraltar	3,548	101.4	+20.6	S\$679.99	0%	
5	SEA	Rotterdam Distripark	1,933	55.2	+27.5	S\$370.51	0%	
6	handoff	Rotterdam Distripark		4.0	+0.0	S\$127.77	3%	
7	ROAD	Port of Rotterdam	31	0.5	+3.6	S\$47.45	0%	
8	handoff	Port of Rotterdam		14.0	+0.0	S\$319.42	8%	

Transit 519.8 h + transfers 18.0 h + announced scenario delay 0.0 h = plan 537.8 h. Model delay buffer p50/p85/p95: +105.6 / +134.4 / +181.5 h. Route band p50 643.4 h, p85 657.4 h, p95 678.6 h (independent legs); 719.4 h if all legs are late together. Costs: linehaul S\$3,531, transfers S\$447.18, disruption surcharge S\$0.00.

Why the slowest leg (CHOKE-BABEL → CHOKE-SUEZ, p85 28.5 h) is slow: exact Shapley attribution of the p85 delay model, hours relative to a 28.9 h baseline



Method: legs are priced with gradient-boosted quantile models (p50/p85/p95) trained on synthetic delays fitted to published UNCTAD, World Bank and STB statistics; FASTEST plans on p50, BALANCED on p85 (plus cargo carrying cost when a value is given), SAFEST on p95 plus 240 risk-hours per unit of threat. Route bands assume independent legs. Sea distances are great-circle between chokepoint waypoints. Currency conversion: live (open.er-api.com) as of Tue, 29 Sep 2026 00:02:31 +0000.